ज्ञानम् अमृतम्

JKLU

PS- I WEEKLY REPORT

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1. Executive Summary

This week, the team successfully completed data preprocessing and built, trained, and evaluated multiple machine learning classification models on the UCI dataset comprising 400 records. The pipeline included data preprocessing, encoding, feature scaling, and an 80:20 train-test split before model training. A total of eight algorithms were implemented and benchmarked. The results are highly promising, with several models achieving perfect classification accuracy. Model performance was evaluated using five metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

2. Dataset Overview

Parameter	Details
Source	UCI Machine Learning Repository
Total Records	400
Task Type	Binary / Multi-class Classification
Training Set	320 records (80%)
Testing Set	80 records (20%)

3. Data Preprocessing Pipeline

Before model training, a structured preprocessing pipeline was applied to ensure data quality and consistency.

3.1 Data Cleaning

- Checked for and handled missing/null values
- Removed duplicate records where applicable
- Verified data types and corrected inconsistencies

3.2 Encoding

- **Label Encoding / One-Hot Encoding** applied to all categorical features to convert them into numerical format suitable for ML algorithms
- Target variable was label-encoded for classification

3.3 Feature Scaling

- **Standard Scaling (Z-score normalization)** applied to continuous numerical features
- Ensured all features are on a uniform scale, particularly important for distance-based models like KNN and SVM

3.4 Train-Test Split

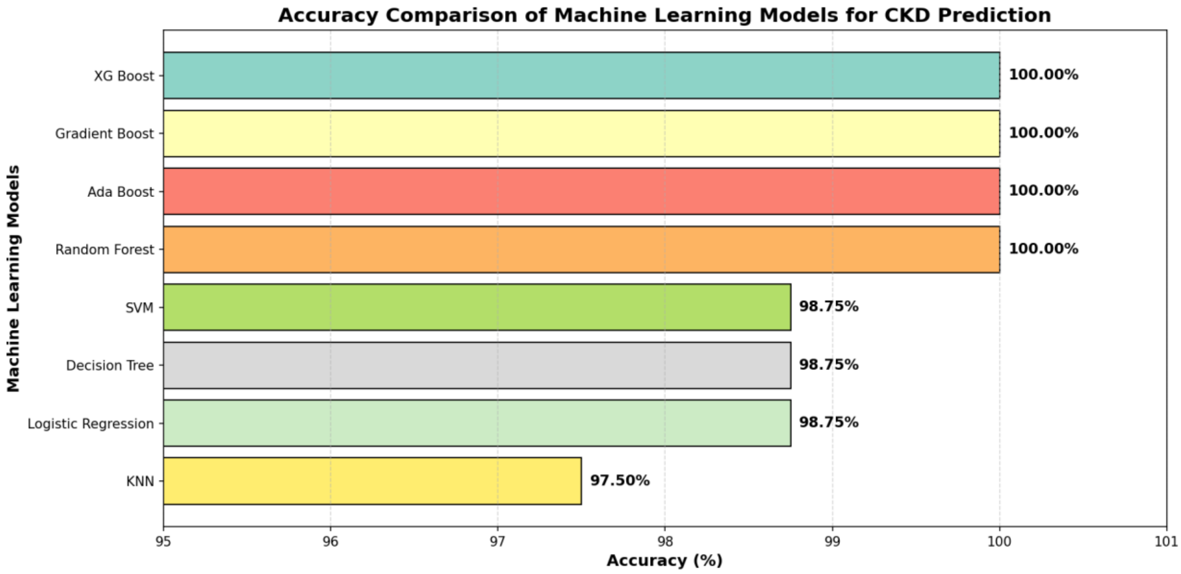
- Dataset split in an **80:20 ratio**
- **Training Set:** 320 records
- **Testing Set:** 80 records

4. Models Implemented & Accuracy Results

Eight classifiers were trained on the preprocessed data. The table below summarizes their accuracy on the test set:

Model	Accuracy	Notes
AdaBoost	100%	Perfect classification
Decision Tree (DT)	98.75%	
Gradient Boosting	100%	Perfect classification
K-Nearest Neighbors (KNN)	97.50%	
Logistic Regression	98.75%	

Model	Accuracy	Notes
Random Forest	100%	Perfect classification
Support Vector Machine (SVM)	98.75%	
XGBoost	100%	Perfect classification



5. Evaluation Metrics

Each model was assessed using the following classification metrics:

Metrics Used:

- **Accuracy** — Overall correct predictions out of total samples
- **Precision** — Of all positive predictions, how many were actually positive
- **Recall** — Of all actual positives, how many were correctly identified
- **F1-Score** — Harmonic mean of Precision and Recall
- **ROC-AUC** — Area under the Receiver Operating Characteristic curve; measures separability

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
AdaBoost	100%	1.00	1.00	1.00	1.00
Decision Tree	98.75%	1.00	0.988	0.989	0.99
Gradient Boosting	100%	1.00	1.00	1.00	1.00
KNN	97.50%	1.00	0.96	0.9796	0.99
Logistic Regression	98.75%	1.00	0.98	0.9899	1.00
Random Forest	100%	1.00	1.00	1.00	1.00
SVM	98.75%	1.00	0.98	0.9899	1.00
XGBoost	100%	1.00	1.00	1.00	1.00

While the results across all models are impressive, **it is important to note that the dataset contains only 400 records**, which is relatively small for drawing definitive conclusions about model performance.

- A small dataset increases the risk of **overfitting**, where a model learns the training data too well and may not generalize to new, unseen data.
- Perfect 100% accuracy on a small dataset does not necessarily reflect real-world performance and should **not be solely relied upon**.
- The high scores may partly be a result of the dataset being too limited in variety and complexity to sufficiently challenge the models.
- **It is strongly recommended to identify and incorporate a larger, more diverse dataset** to re-evaluate these models and validate their true generalization ability before any production or deployment consideration.
- Cross-validation and external benchmarking on a separate dataset will be prioritized in the next phase of the project.